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Aligning explanations with human values: context-sensitive trust calibration in AI decision systems

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ABSTRACT

Artificial intelligence (AI) systems increasingly make decisions that require human trust, yet little is known about how users evaluate these decisions when moral reasoning is involved. This study examines how different AI reasoning styles – utilitarian, deontological, and virtue-based reasoning – shape user trust and agreement with AI decisions across everyday, corporate, environmental, and medical contexts. In a controlled experiment ($N = 160$), participants rated AI decisions before and after viewing the system's explanations. Quantitative analysis revealed that explanations significantly increased trust and agreement, with deontological reasoning yielding the largest gains in higher-stake contexts, while utilitarian reasoning showed weaker effects in medical scenarios. A mediation analysis indicated that perceived transparency partially explained the trust increase, with the remainder attributable to value alignment. We propose a context-sensitive framework for designing explanations that calibrate trust by matching reasoning style to domain expectations and decision stakes.

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Trust calibration; explainable AI; value alignment; reasoning styles; human – AI interaction

1. Introduction

Artificial intelligence (AI) is no longer a purely technical endeavour – designers and researchers in human – computer interaction (HCI) and information systems (IS) increasingly grapple with how to build systems that users can appropriately rely upon. Early research on automation warned that people often miscalibrate trust, either over – or under-relying on algorithmic advice, and argued that trust should be treated as a design variable rather than a purely moral judgement (Kizilcec 2016). Subsequent work in IS distinguished between interpersonal trust and trust in technology, highlighting that the latter involves beliefs about a system's competence, reliability and benevolence (McKnight, Carter, and Clay 2009). Trust building is thus a dynamic process shaped by human, environmental and technological factors, moving from initial formation to ongoing development.

Within HCI, recent comparative studies show that explanation methods have heterogeneous effects on users' understanding, uncertainty recognition and calibrated trust (Wang and Yin 2021). Many popular explanation techniques fail to improve calibrated trust when users have little domain expertise, particularly when explanations are overly complex or fail to match users'

mental models (Wang and Yin 2021). These insights resonate with trust calibration research in automation, which stresses that explanations should help people build appropriate reliance rather than blind acceptance (Kizilcec 2016). Our work positions itself at this intersection, asking how explanations can be designed to align with human values and context to foster calibrated trust.

AI systems are increasingly expected to justify their decisions, yet not all explanations equally enhance user trust. This puzzle becomes particularly acute as AI assumes greater responsibility for consequential decisions – from medical diagnoses to criminal justice assessments. Such variability is unsurprising when moral acceptability depends on contextual alignment between AI reasoning and human moral expectations (Van Berkel et al. 2022). While the demand for transparent, explainable AI has intensified, emerging evidence reveals a paradoxical landscape where explanations sometimes build trust, sometimes destroy it, and sometimes have no effect at all. Consider what happens when users encounter an AI's decision that surprises them. In high-stakes educational contexts, algorithmic transparency demonstrably increases trust when grading outcomes violate student expectations – the explanation

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essentially buffers the shock by revealing a fair, systematic process (Kizilcec 2016). This protective effect extends broadly across domains: in government services, citizens who receive any form of explanation from algorithmic recommender systems report significantly higher trust and greater willingness to follow AI recommendations than those given no justification at all (Nieuwenhuizen et al. 2025). The mere act of explaining appears to signal accountability and good faith, encouraging user acceptance.

Yet this seemingly straightforward relationship quickly unravels under scrutiny. Clinical decision support systems reveal how the same transparency that builds trust in one context can undermine it in another – certain explanation types lead to dangerous overtrust in flawed systems, while others create confusion that erodes confidence entirely. Even more troubling, providing excessive detail about an algorithm’s inner workings can backfire, with information overload destroying the very trust that moderate transparency creates (Kizilcec 2016). These contradictions suggest that the value of explanations depends critically on factors we’re only beginning to understand.

The value dimension of AI decisions introduces another layer of complexity that existing explanation research has largely overlooked. Recent experiments reveal that people fundamentally distrust AI systems that employ utilitarian reasoning – maximising overall outcomes at the expense of individual rights – even when they agree with the specific decisions (Myers and Everett 2025). This aversion persists regardless of whether the advisor is human or artificial, but intensifies for AI systems perceived as lacking warmth and empathy (McKee, Bai, and Fiske 2023). When an AI agent knowingly chooses a risky action that prioritises mission success over safety – a utilitarian trade-off – explanations only partially repair trust, failing to restore perceptions of benevolence or integrity. The AI reasoning style underlying a decision thus appears fundamental to user acceptance, yet we know little about how different decision logics interact with transparency efforts.

Context shapes these dynamics in profound ways. When users have high personal stakes in a decision, algorithmic transparency produces dramatically stronger trust gains than in low-involvement situations, suggesting that people scrutinise explanations more carefully when outcomes matter to them personally. This pattern aligns with dual-process models of persuasion, where motivated users engage in deep, systematic processing of information rather than relying on superficial cues. Domain characteristics further modulate trust: medical contexts exhibit stronger algorithm aversion than everyday decision scenarios, while

corporate and environmental decisions occupy a middle ground where value considerations loom large but technical expertise remains accessible. Evidence from personalised medicine likewise shows that trust strongly moderates the adoption of AI-driven innovations in high-stake healthcare settings (Misra, Bisui, and Singh 2020). These scattered findings point toward an unexplored interaction between value reasoning styles, contextual factors and explanation effectiveness. When do explanations build trust rather than destroy it? Why do some AI reasoning styles make decisions more acceptable than others? How do contextual stakes shape users’ receptivity to different types of justifications? Current research has examined these dimensions in isolation, missing the complex interplay that determines real-world trust outcomes. This interplay highlights that transparency, authority cues, and contextual value alignment jointly shape user trust (Cetinkaya and Krämer 2025; Ha et al. 2022).

This study addresses these gaps through three key contributions. First, we present empirical evidence from a controlled experiment ($N = 160$) systematically varying AI reasoning styles (utilitarian, virtue ethics, deontological), decision contexts (medical, corporate, environmental, everyday) and explanation presence. Second, we develop a theoretical framework integrating a Dual-Process Model of Value Alignment to explain when and why explanations build trust – the Context – Value Alignment Model (CVAM) provides a unified account of how value considerations and information processing dynamics jointly determine user responses. Our model explains how users evaluate AI decisions based on the alignment between the AI’s reasoning style and their own value expectations. Third, we derive practical design principles for developing context-sensitive explanation systems that align with users’ value expectations and decision stakes, moving beyond one-size-fits-all transparency toward selective, value-aware explanation strategies. Our central research question is: *How do different AI reasoning styles and contextual stakes shape the effectiveness of AI reasoning explanations on user trust and agreement?*

The following sections present our theoretical framework, experimental methodology and empirical findings. We conclude by discussing the implications for both AI system design and our broader understanding of trust in algorithmic decision-making. As AI continues to permeate critical domains of human life, understanding the nuanced relationship between explanations and trust becomes not just an academic interest but a practical imperative for responsible AI deployment.

1.1. Trust and value alignment in explanation design

We distinguish between ethical frameworks and AI reasoning styles: Ethical frameworks refer to normative theories from moral philosophy that guide moral decision-making – specifically utilitarianism (maximising aggregate outcomes), deontology (adhering to rules and duties), and virtue ethics (prioritising moral character and wisdom).

AI reasoning styles refer to the computational implementation of these frameworks in algorithmic decision-making systems. Our experimental AI agents employ reasoning styles grounded in these ethical frameworks but adapted for practical AI decision logic.

Throughout this paper, we use ‘ethical frameworks’ when discussing philosophical foundations and prior research, and ‘AI reasoning styles’ when describing our experimental manipulation and AI system behaviour. This distinction recognises that while AI systems can implement decision logics inspired by moral philosophy, their computational reasoning may differ from human moral deliberation. Throughout this paper, value alignment refers to the perceived congruence between an AI system’s decision logic and the user’s moral intuitions and expectations. It is distinct from mere comprehension of AI reasoning – a user may understand a utilitarian calculation without feeling that it reflects their values. Operationally, value alignment is captured in this study through trust and agreement items that explicitly tap into value congruence, including ‘How much do you feel the AI’s decision-making aligns with your personal values?’ (Table A1). This construct bridges the cognitive dimension of explanation understanding and the affective dimension of moral acceptance.

1.2. Explanation and trust in AI

The quest to make AI systems trustworthy through transparency has produced a rich but contradictory body of evidence. At its core, transparency operates as a form of procedural justice – when users understand how decisions are made, they’re more likely to perceive them as fair and legitimate. This principle manifests clearly in algorithmic grading systems, where students who receive unexpectedly low grades lose trust unless the system explains its reasoning, effectively buffering the disappointment by demonstrating systematic fairness (Kizilcec 2016). The explanation transforms an opaque verdict into a comprehensible process, allowing users to evaluate whether they were treated fairly even if they dislike the outcome. This transparency effect

extends across domains and explanation types. Government algorithmic systems that provide any form of explanation – whether describing the decision process, offering rationales, or giving directive guidance – consistently earn greater citizen trust than their opaque counterparts (Nieuwenhuizen et al. 2025). The universality of this finding suggests that explanations fulfil a fundamental psychological need: users want to know that consequential decisions affecting their lives follow comprehensible logic rather than arbitrary black-box processes. Even in commercial recommender systems, different explanation facilities enhance distinct dimensions of trust – ‘how’ explanations boost perceived competence, ‘why’ explanations increase benevolence, and trade-off explanations strengthen integrity perceptions (Wang and Benbasat 2007). Yet the relationship between transparency and trust follows a dangerous curve. Initial increases in algorithmic transparency reliably boost user confidence, but excessive detail triggers a precipitous decline – users become overwhelmed by information they cannot process, leading to confusion and renewed distrust (Kizilcec 2016). This inverted-U relationship reveals the delicate balance required: enough transparency to establish understanding, but not so much that it creates cognitive overload. Recent work shows that transparency and trust are interlinked through users’ integrity, ability, and benevolence judgments, emphasising that perceived quality of transparency cues drives trust formation (Cetinkaya and Krämer 2025).

The type of explanation matters as much as its presence. In clinical decision support, different explanation classes produce wildly different outcomes – counterfactual explanations can build calibrated trust by helping users understand decision boundaries, while global rule descriptions often overwhelm and confuse, leading to dangerous miscalibration where users either overtrust flawed systems or undertrust accurate ones. Perceived authority of the AI itself can further shape these perceptions, as higher-status systems offering explanations are often viewed as more reliable and controllable (Ha et al. 2022). Recent studies in HCI/IS further show that explanation methods do not universally improve calibrated trust; some approaches may actually mislead users or mask model uncertainty (Wang and Yin 2021). Appropriate reliance therefore requires not only making AI reasoning visible but designing explanations that support users’ mental models (Kizilcec 2016).

1.3. Value reasoning and trust

The value dimension of AI decisions introduces complexities that pure transparency cannot resolve.

Extensive experimental evidence reveals that people exhibit profound algorithm aversion when AI systems make value-laden decisions, trusting them significantly less than human decision-makers even when their advice is identical (Myers and Everett 2025). This aversion intensifies dramatically when AI systems employ utilitarian reasoning that maximises aggregate outcomes while accepting individual harm. Users expect AI advisors to follow utilitarian logic – treating decisions as mathematical optimisation problems – yet they paradoxically distrust systems that actually reason this way, creating a fundamental tension between expectations and acceptance. People consistently view AI systems as lacking warmth and empathy, qualities deemed essential for moral judgement (McKee, Bai, and Fiske 2023). When forced to evaluate value violations, users apply different standards to AI versus human actors – the same transgression that would trigger moral outrage against a human barely registers when committed by an AI, not because the harm is less real but because users doubt the AI’s capacity for genuine value reasoning (Bigman and Gray 2018). This perception creates a double bind: AI systems are simultaneously seen as too calculating to make genuinely moral decisions and too mechanical to be held accountable for their choices.

The decision logic underlying AI reasoning profoundly shapes trust outcomes. Deontological approaches that emphasise rules, duties, and individual rights align more naturally with human value intuitions than utilitarian calculations. When an AI agent must explain why it knowingly chose a risky action that prioritised mission success over safety – a classic utilitarian trade-off – even detailed explanations fail to fully restore trust, particularly regarding the system’s perceived benevolence and integrity. Users can understand and even accept the logic behind utilitarian decisions, but this intellectual understanding doesn’t translate into emotional trust. They want AI systems to respect inviolable principles, not just optimise outcomes.

1.4. Context and stakes

The context surrounding AI decisions fundamentally alters how users process and respond to explanations. High-involvement situations – where outcomes directly affect users’ wellbeing or core interests – trigger fundamentally different cognitive processes than low-stakes decisions. When personally invested in outcomes, users exhibit dramatically stronger responses to algorithmic transparency, with trust gains several times larger than in routine contexts. This differential effect suggests that explanations don’t simply provide

information; they serve different psychological functions depending on what’s at stake. Domain characteristics create another layer of contextual variation. Medical decisions trigger the strongest algorithm aversion, with users preferring human judgement even when AI demonstrably outperforms human experts. This resistance persists across medical subfields but varies in intensity – AI psychiatric advice faces near-universal rejection while diagnostic imaging applications earn grudging acceptance. The pattern extends beyond healthcare: after witnessing a single error, users abandon algorithmic forecasters more readily than human ones making identical mistakes, even when the algorithm maintains superior overall accuracy (Dietvorst, Simmons, and Massey 2015). These domain-specific trust patterns suggest that explanations must overcome different baseline scepticism levels depending on the decision context.

1.5. Trust calibration and backfire effects

The ultimate goal of explainable AI isn’t simply to maximise trust but to calibrate it appropriately – users should rely on AI when it’s accurate and maintain healthy scepticism when it’s not. Yet explanations often miscalibrate rather than calibrate trust. Systematic reviews of explainable AI in healthcare reveal a troubling pattern: while most studies find explanations enhance trust, a significant minority report decreased trust when explanations are overly complex, incoherent or reveal reasoning that contradicts user expectations. Moreover, the very act of asking participants to rate trust can unintentionally alter their subsequent reliance on AI recommendations, a ‘question – behaviour’ effect that complicates trust-calibration research (Schrills et al. 2025). The explanation that confuses or alarms can be worse than no explanation at all. This backfire risk becomes particularly acute when explanations expose misalignment between AI reasoning and user values. Detailed transparency about an algorithm’s utilitarian calculations might technically increase understanding while simultaneously decreasing trust by highlighting the system’s cold decision logic. The challenge isn’t just making AI decisions comprehensible but making them acceptable. Both overtrust in flawed systems and undertrust in accurate ones produce harmful outcomes, yet current explanation approaches often push users toward these extremes rather than appropriate middle ground (Wang and Yin 2021).

Trust repair mechanisms offer partial solutions but come with their own complexities. When AI systems fail or make ethically questionable decisions, well-crafted apologies can partially restore user confidence,

particularly when they accept responsibility and demonstrate understanding of the harm caused (Kulms and Kopp 2018). However, trust violations stemming from deliberate value trade-offs – where the AI knowingly chose a course that violated user expectations – prove far more difficult to repair than simple errors. An explanation that reveals intentional violation of user values may permanently damage trust regardless of how clearly it's communicated.

1.6. Integrating the dual-process model of value alignment

Making sense of these varied findings requires theoretical frameworks that can accommodate both value and cognitive dimensions of trust formation. We integrate Moral Foundations Theory and the Elaboration Likelihood Model into a Dual-Process Model of Value Alignment. Moral Foundations Theory provides a lens for understanding how users evaluate AI decisions through innate moral intuitions organised around care/harm, fairness/cheating, loyalty/betrayal, authority/subversion and purity/sanctity. AI decisions that violate these foundations – particularly care and fairness – trigger immediate moral resistance regardless of their logical validity. Users judge identical moral violations differently when committed by AI versus humans, suggesting that AI actions are processed through a distinct value framework that emphasises different foundations (Haidt and Graham 2007). The Elaboration Likelihood Model (ELM) explains how users process explanations under different conditions. When motivation and ability are high – as in high-stakes moral decisions – users follow the central route, carefully scrutinising explanation quality and logical coherence. The strength of value arguments becomes paramount, and superficial features matter less. In low-involvement contexts, users default to peripheral processing, relying on heuristic cues like source credibility or presentation style rather than engaging deeply with explanation content (Petty and Cacioppo 1986). This synthesis reveals why explanation effectiveness varies across contexts: high-stakes decisions involving value foundations trigger central-route processing, making users highly sensitive to whether the AI's reasoning respects moral boundaries or merely optimises outcomes, whereas low-stakes decisions render elaborate explanations superfluous. Effective explanation strategies must simultaneously address value alignment and match users' cognitive processing modes – a challenge that requires moving beyond universal transparency toward context-sensitive, value-aware explanation design.

1.7. Model of context-sensitive explanation effectiveness

The theoretical landscape we've traversed reveals a complex interplay between explanations, value reasoning and contextual factors that demands an integrated framework. Our Context – Value Alignment Model (CVAM) emerges from this synthesis, proposing that explanation effectiveness isn't simply about transparency – it's about achieving alignment between the value logic revealed by explanations and the cognitive – emotional processing triggered by contextual stakes.

At its foundation, the model posits a direct pathway from transparency to trust: when AI systems explain their reasoning, users gain insight into decision processes that would otherwise remain opaque, fostering confidence through understanding. But this direct relationship tells only part of the story. The real power of explanations operates through two critical mediating mechanisms. First, explanations enhance perceived fairness by demonstrating that decisions follow systematic principles rather than arbitrary calculations – users need to see not just what the AI decided, but that it decided fairly. Second, and more subtly, explanations reveal the value framework underlying AI reasoning, allowing users to assess whether the system's values align with their own intuitions. When an AI explanation shows adherence to deontological principles – respecting individual rights, following inviolable rules – it signals moral integrity that resonates with users' fairness and care foundations.

Context fundamentally moderates these relationships. In high-stakes situations – medical decisions affecting health, corporate choices impacting livelihoods, environmental judgements shaping collective futures – users shift into central-route processing, carefully scrutinising both the logical coherence and value acceptability of explanations. The stakes activate deeper engagement, making users sensitive to whether the AI's reasoning respects moral boundaries or merely optimises outcomes. Conversely, everyday decisions trigger peripheral processing where elaborate moral justifications become unnecessary cognitive burdens. The model thus predicts that contextual risk doesn't just amplify the effect of explanations uniformly; it specifically enhances the importance of value alignment between AI reasoning and user expectations. This framework challenges conventional assumptions about user characteristics. While existing literature often emphasises demographic differences in AI acceptance, our model suggests that value and contextual factors overwhelm individual differences. The universal nature of moral foundations and the situational activation of processing routes should produce consistent patterns across

diverse user populations – a claim that directly contradicts personalization-focused approaches to explainable AI.

From this conceptual model flow five testable hypotheses that capture the key relationships:

H1: Providing reasoning explanations increases user trust and agreement with AI decisions. This foundational hypothesis establishes that transparency generally benefits trust, consistent with procedural justice principles and the universal human need to understand consequential decisions affecting our lives.

H2: The effect of explanations on trust is strongest for AI decisions using deontological reasoning, moderate for virtue ethics reasoning, and weakest for utilitarian reasoning. Different AI reasoning styles trigger varying levels of alignment with human value intuitions. Deontological reasoning's emphasis on rules and rights should resonate most strongly when made transparent, while utilitarian calculations may actually undermine trust when exposed.

H3: Perceived transparency mediates the relationship between explanations and trust. Beyond direct effects, explanations operate by enhancing users' subjective sense of understanding and insight into AI decision-making, which in turn builds confidence in the system's trustworthiness.

H4: Contextual risk amplifies the positive effect of explanations on trust, with stronger benefits in high-stakes (medical, corporate, environmental) versus low-stakes (everyday) contexts. Following ELM predictions, high-stakes contexts trigger central-route processing that makes users more responsive to explanation quality and value alignment, while low-stakes situations reduce the marginal value of detailed justifications.

H5: User demographic factors (age, education, AI proficiency) do not significantly moderate the explanation – trust relationship. The universal nature of moral processing and situational determinants of cognitive engagement should produce consistent effects across user populations, challenging assumptions about demographic segmentation in XAI design.

These hypotheses collectively test whether effective AI explanations must go beyond mere transparency to achieve value and contextual alignment. If supported, they would fundamentally reframe explainable AI from a technical challenge of making algorithms interpretable to a design challenge of crafting explanations that respect both human values and situational demands.

2. Methods

2.1. Study design and participants

The study employed a mixed experimental design to examine how reasoning explanations affect user trust and agreement with AI decisions across different AI

reasoning styles and contextual scenarios. The independent variables were Explanation Presence (no reasoning vs. reasoning shown), AI Reasoning Style (utilitarianism, deontology, virtue ethics), and Scenario Context (everyday life, environmental ethics, corporate ethics, medical emergency). The four contextual domains were selected to represent a theoretically motivated range of decision stakes and value salience. Everyday life scenarios served as a low-stakes baseline where peripheral processing is expected to dominate. Environmental and corporate ethics scenarios represent moderate-to-high stakes contexts characterised by competing stakeholder interests and value trade-offs (Van Berkel et al. 2022). Medical emergency scenarios represent the highest-stakes domain, where algorithm aversion is most pronounced in prior literature (Dietvorst, Simmons, and Massey 2015) and utilitarian calculations most sharply conflict with sanctity-of-life intuitions. The dependent variables were Trust in AI's decision and Agreement with AI's decision, both measured on 7-point Likert scales (1 = 'not at all,' 7 = 'completely'). The experiment consisted of two groups: Group 1 ($n = 100$) rated AI-generated value decisions without seeing reasoning, while Group 2 ($n = 60$) rated the same AI decisions before and after viewing the AI's reasoning process, allowing within-subject comparison. This design enabled both between-group (with vs. without explanation) and within-group (pre – vs. post-reasoning) comparisons, maximising statistical power while controlling for individual differences. The unequal group sizes arise from the differing analytical roles of the two experimental conditions. Group 2 employed a within-subject pre – post design, which affords higher statistical sensitivity per participant but also has greater cognitive and temporal demands due to repeated evaluations. Accordingly, a smaller sample was assigned to this condition, while a larger between-subject group (Group 1) was used to establish baseline trust and agreement in the absence of explanations.

A total of 160 participants (ages 19–65) were recruited through online portals and participant recruitment platforms. Participation was voluntary and no monetary compensation was provided. To ensure data quality, attention checks were embedded within the task (e.g. items instructing participants to select a specific response to confirm attentive engagement). No prior topic-specific familiarity was screened; instead, all scenarios were designed to be interpretable by lay participants and were presented with sufficient contextual information to support meaningful evaluation though familiarity with digital interfaces was screened to ensure basic interaction ability. The sample included 87 males (54%), 55 females (34%), and 18 participants

(11%) who chose not to disclose gender. Participants' technological proficiency varied: 36% beginners, 51% intermediate, and 13% advanced. All participants gave informed consent prior to participation.

2.2. Materials and procedure

Three AI agents representing distinct AI reasoning styles were implemented using prompt engineering with GPT-4. No model fine-tuning or additional training was performed. Each agent followed a fixed system-level prompt that operationalised a specific reasoning style (utilitarian, deontological, or virtue-based), as detailed in Appendix A (Table A2 and A3). The utilitarian agent maximised aggregate benefit through outcome-based logic, the deontological agent followed fixed moral rules and duties through rule-based logic, and the virtue ethics agent prioritised moral character and empathy in decision-making. Each agent was applied to four ethical scenarios representing different contextual domains: everyday life dilemmas involving personal moral choices, environmental ethics balancing sustainability against convenience, corporate ethics weighing profit against fairness, and medical emergencies requiring decisions between saving one versus many lives. All AI responses were pre-generated to ensure consistency, with each scenario containing three AI-generated decisions (one per reasoning style) presented to participants in randomised order.

Participants completed the experiment online through a browser-based interface. Group 1 viewed each scenario and rated trust and agreement using multi-item 7-point Likert scales (see Appendix Table A1). Group 2 followed a two-stage process. In Stage 1 (Pre-Reasoning), participants rated trust and agreement before seeing the AI's reasoning. In Stage 2 (Post-Reasoning), participants viewed the AI's reasoning explanation (2–3 sentences explaining its decision logic) and then re-rated trust and agreement.

Each participant evaluated all four scenarios. The ethical frameworks and reasoning order were randomised. A 5-minute break was included between the pre- and post-reasoning phases to reduce carryover effects. After completing all tasks, participants filled out a short transparency perception scale (e.g. 'The AI's reasoning was clear to me,' rated on a 1–7 Likert scale) and a demographic questionnaire. To ensure reproducibility and consistency across participants, all AI decisions and explanations were pre-generated using fixed prompt templates. The system-level prompt structure for each reasoning style is reported in Appendix A (Table A2), and illustrative examples of scenario

prompts and corresponding AI outputs are provided in Appendix A (Table A3).

2.3. Measures

Trust and agreement served as the primary dependent measures, while perceived transparency was examined as a mediator, and the remaining variables as potential moderators. Table 1 below presents the study measures and the relevant coding.

2.4. Data analysis

Analyses were conducted using Python and R. Paired-samples *t*-tests were used to compare pre- versus post-reasoning trust and agreement ratings within Group 2, while independent-samples *t*-tests compared post-reasoning trust and agreement between Group 1 (no explanation) and Group 2 (with explanation). Mixed-effects models with random intercepts for participants were employed to examine interactions between AI reasoning style and scenario context, accounting for the repeated-measures structure of the data.

Mediation analysis was conducted to test whether perceived transparency explained the relationship between explanation presence and trust. The indirect effect of transparency was significant ($b = 0.105$, 95% CI [0.04, 0.17]), indicating partial mediation. Ordinal logistic regression models were additionally estimated to examine the robustness of trust effects across ordered response categories, yielding increased odds of higher trust ratings in the explanation condition (OR ≈ 2.35).

No participants were excluded from the analysis, and no missing responses were observed. All Likert-scale responses were analysed as collected, with trust and agreement treated as multi-item measures (composite scores formed by aggregating their respective Likert

Table 1. Study measures and scales.

Variable	Description	Scale/Coding
Trust in AI	Perceived trustworthiness of AI's decision	1–7 Likert
Agreement	Degree of agreement with AI's decision	1–7 Likert
Perceived Transparency	Perceived clarity and openness of AI reasoning	1–7 Likert
Prior AI Familiarity	Self-rated familiarity with AI tools	0–2 (low, medium, high)
Adoption Tendency	Self-rated willingness to adopt AI systems	0–2 (low, medium, high)
Technical Proficiency	Self-rated skill in using digital tools	0–2 (low, medium, high)
General Trust in Technology	Baseline trust orientation	1–7 Likert
Age	Continuous variable	Years

items). Given the ordinal nature of the response scales and the repeated-measures design, robustness checks were conducted to assess whether the reported effects were dependent on parametric assumptions. Mann – Whitney U tests confirmed that trust was significantly higher in the explanation condition than in the no-explanation condition ($U = 355070.00$, $p < .001$), with a corresponding but smaller effect for agreement ($U = 376061.00$, $p < .001$). Permutation tests (10,000 iterations) yielded convergent results: explanations increased mean trust by 0.43 points ($p < .001$; standardised effect $\approx d = 0.32$) and mean agreement by 0.31 points ($p < .001$; standardised effect $\approx d = 0.23$). Median-based permutation tests were weaker for agreement and did not reach significance, reflecting the coarse granularity of Likert scales and clustering around identical median values rather than a contradiction of the mean-based effects.

The intraclass correlation coefficient ($ICC = 0.37$) indicated substantial participant-level clustering, justifying the use of mixed-effects modelling. All significance tests used $\alpha = 0.05$. Power considerations indicated that a sample size exceeding 150 participants was sufficient to detect medium-sized effects (Cohen's $d \approx 0.5$), a threshold exceeded by the final sample ($N = 160$).

A detailed summary of all statistical analysis is provided in Appendix A (Table A4).

2.5. Ethical statement

All research procedures and studies on humans (individuals, sample, or data) were performed in accordance with the principles stated in the Declaration of Helsinki.

3. Results

3.1. Main effects: do explanations build trust?

Our central question – whether AI explanations enhance user trust – required first establishing a baseline effect before exploring nuances. Across all conditions, providing reasoning explanations significantly increased trust compared to unexplained decisions ($t(158) = 5.24$, $p < .001$, $d = 0.426$). This represents a meaningful shift: trust scores rose from 4.39–4.82 on our 7-point scale, moving users from mild scepticism toward moderate confidence. Agreement showed parallel gains, increasing by 0.31 points ($t(158) = 3.79$, $p < .001$, $d = 0.308$), suggesting that understanding an AI's reasoning helps users both trust the system and align with its conclusions. Table 2 below reports the overall effect of explanations on trust and agreement.

Table 2. Effect of explanations on trust and agreement ($N = 160$).

Measure	No Explanation M (SD)	With Explanation M (SD)	t	p	d
Trust	4.39 (1.42)	4.82 (1.31)	5.24	<.001	0.426
Agreement	4.51 (1.38)	4.82 (1.29)	3.79	<.001	0.308

Within-subject comparisons in Group 2 strengthened this finding. When the same participants evaluated decisions before and after seeing explanations, trust jumped immediately (pre: $M = 4.41$, post: $M = 4.84$, paired $t(59) = 4.88$, $p < .001$). This within-person shift eliminates individual difference confounds, confirming that explanations causally influence trust rather than merely correlating with trusting personalities.

3.2. Reasoning style effects: not all decision logics are equal

Having established that explanations help overall, we tested whether the AI reasoning style underlying the decision matters (H2). The answer was unequivocal: reasoning style dramatically shaped both baseline trust and explanation effectiveness. Mixed-effects models revealed a significant Reasoning Style main effect ($F(2, 1916) = 47.83$, $p < .001$) and crucially, a Reasoning Style $\times$ Explanation interaction ($F(2, 1916) = 7.21$, $p < .001$). A summary of trust scores by AI reasoning style and explanation condition is presented in Table 3 below.

As Figure 1 illustrates, deontological explanations – those emphasising rules and duties – generated the highest absolute trust ($M = 5.21$) and the largest gains from transparency (increase of 0.52 points). Virtue ethics occupied middle ground, while utilitarian reasoning, despite its logical clarity, produced both the lowest trust and smallest benefits from explanation. Post-hoc comparisons confirmed these differences were not just statistically significant but practically meaningful (all $p < .01$, Bonferroni corrected). The pattern suggests users do not simply want to understand AI decisions – they want those decisions grounded in value principles they share.

3.3. Context matters: where explanations work best

We next examined whether scenario context modulates explanation effectiveness (H4). Baseline trust varied

Table 3. Trust scores by AI reasoning style and explanation condition.

Reasoning Style	No Explanation M (SD)	With Explanation M (SD)	Difference
Deontological	4.69 (1.32)	5.21 (1.18)	0.52
Virtue Ethics	4.36 (1.41)	4.77 (1.29)	0.41
Utilitarian	4.02 (1.51)	4.31 (1.45)	0.29

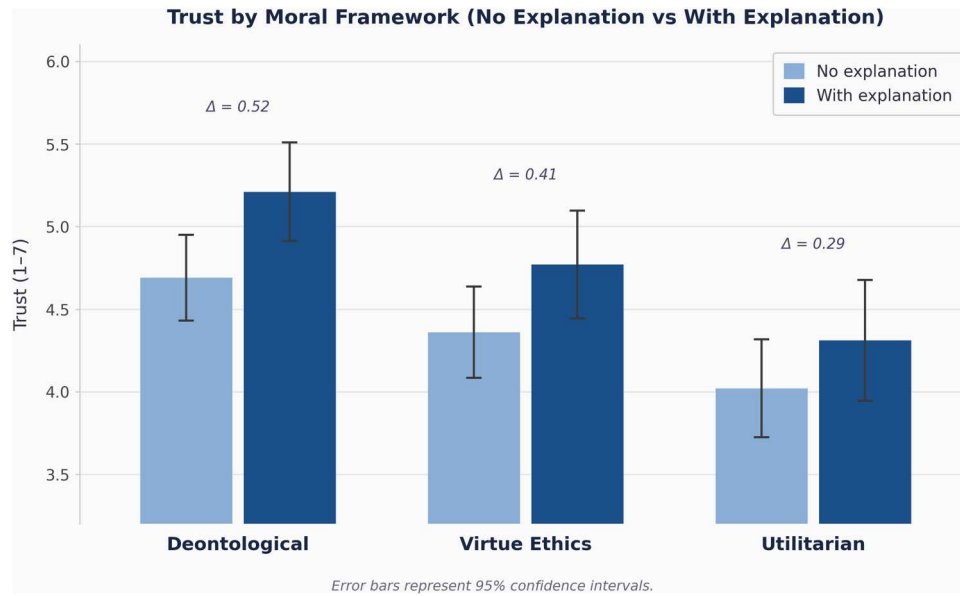


Figure 1. Trust scores by AI reasoning style and explanation condition. Error bar represent 95% confidence intervals.

significantly across domains ($F(3, 1914) = 19.64$, $p < .001$), with everyday scenarios earning highest trust ($M = 5.11$) and medical contexts triggering strongest scepticism ($M = 4.45$). However, the magnitude of explanation benefits followed a different pattern entirely. Table 4 reports trust and explanation effects across the four scenario contexts.

Corporate and environmental contexts showed the strongest explanation benefits (0.48 and 0.45 respectively), despite moderate baseline trust. These domains – characterised by value trade-offs and stakeholder tensions – apparently benefit most from transparency about decision logic. Everyday scenarios, already trusted, gained less from explanations (0.32), while medical contexts showed intermediate effects (0.37) despite lowest baseline trust, suggesting that domain-specific algorithm aversion can limit explanation benefits.

3.4. The critical interaction: when AI reasoning meets context

The Reasoning Style $\times$ Scenario interaction revealed our most nuanced finding ($F(6, 1910) = 3.21$, $p = .004$). As

shown in Figure 2, deontological explanations proved especially powerful in corporate and environmental contexts, generating trust increases of 0.75 and 0.71 points – nearly double the main effect.

Table 5 shows explanation effect sizes (Cohen's d) by AI reasoning style and scenario.

Conversely, utilitarian explanations barely helped in medical contexts ($d = 0.18$), where calculating lives saved versus lost apparently violates deep value intuitions about medical ethics. This interaction pattern confirms that effective explanations must achieve alignment between AI reasoning style and contextual expectations.

3.5. Understanding the mechanism: the role of transparency

To understand how explanations build trust, we tested Perceived Transparency as a mediator (H3). As illustrated in Figure 3, Perceived Transparency partially mediated the relationship, with a significant indirect effect ($b = 0.105$, 95% CI [0.04, 0.17]). However, this pathway explained only 20% of the total effect – the direct effect remained substantial ($b = 0.421$, $p < .001$).

Table 6 reports the total, direct, and indirect effects from the mediation analysis.

This partial mediation suggests that while understanding how an AI thinks matters, explanations build trust through additional mechanisms – likely including perceptions of value alignment, procedural fairness, and institutional accountability that transcend mere comprehension.

Table 4. Trust and explanation effects across scenario contexts.

Scenario	Overall Trust M (SD)	Explanation Effect
Everyday	5.11 (1.09)	0.32
Environmental	4.73 (1.27)	0.45
Corporate	4.58 (1.34)	0.48
Medical	4.45 (1.48)	0.37

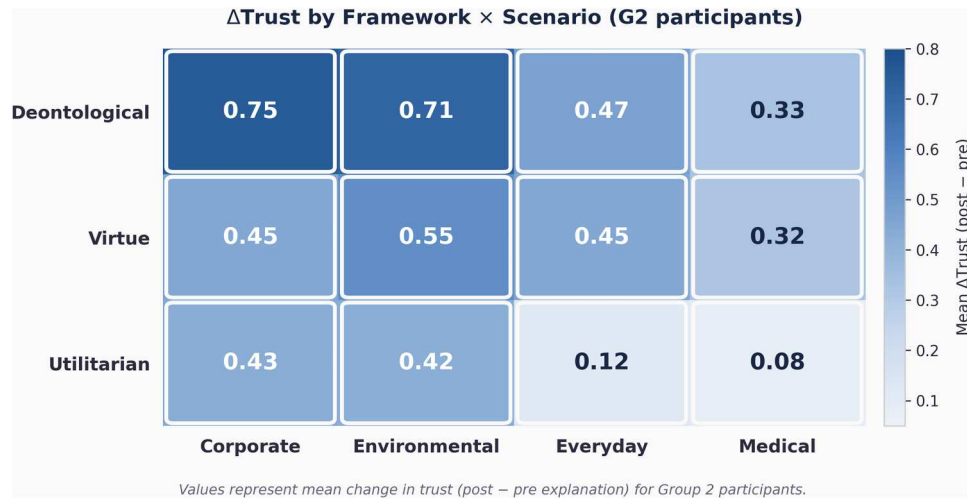


Figure 2. Interaction between AI reasoning style and scenario context on explanation effectiveness. Values represent change in trust with explanations.

Table 5. Explanation effect sizes (Δ trust) by AI reasoning style and scenario.

Reasoning Style	Medical	Corporate	Environmental	Everyday
Deontological	0.43	0.75	0.71	0.38
Virtue Ethics	0.39	0.42	0.44	0.35
Utilitarian	0.18	0.31	0.35	0.28

3.6. Robustness: do individual differences matter?

Contrary to H5's prediction, demographic factors showed remarkably little influence. Neither age ($p = .43$), education ($p = .61$) nor AI proficiency ($p = .38$) moderated explanation effectiveness. Even technology novices benefited as much from explanations as experts. Only baseline trust in technology showed weak moderation ($b = -0.08, p = .03$), with low-trust individuals gaining slightly more from explanations – a ceiling effect rather than meaningful heterogeneity. These null moderations held across multiple analytical approaches. Ordinal logistic regression treating trust as ordered categories yielded $OR = 2.35$ (95%

CI [1.74, 3.18]) for explanations. Non-parametric Wilcoxon tests confirmed all findings without distributional assumptions. Bootstrap confidence intervals (1000 iterations) remained stable, and the intraclass correlation of 0.37 justified our mixed-effects approach while revealing that trust varied more within individuals (across decisions) than between them.

Together, these results tell a clear story: explanations generally increase trust in AI decisions, but their effectiveness depends critically on value alignment and contextual fit. Deontological reasoning resonates most strongly, especially in domains requiring principled judgement. While transparency mediates some of this effect, the majority operates through other pathways – likely involving perceived value alignment and procedural justice. Individual differences matter surprisingly little; value and contextual factors dominate. These patterns held across multiple robustness checks, suggesting they reflect genuine psychological phenomena rather than analytical artefacts. We next consider what these findings mean for theory and practice.

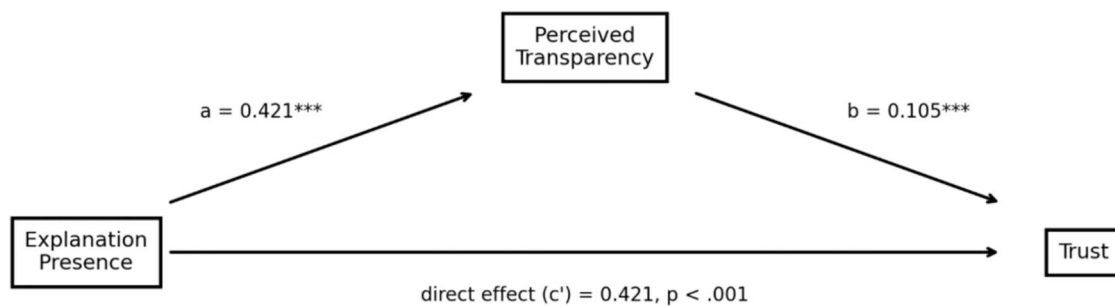


Figure 3. Mediation model showing direct and indirect effects of explanations on trust through perceived transparency. Values represent unstandardised coefficients. *** $p < .001$.

Table 6. Mediation analysis: explanation → perceived transparency → trust.

Path	Coefficient	SE	95% CI
Total Effect	0.526	0.087	[0.355, 0.697]
Direct Effect	0.421	0.079	[0.266, 0.576]
Indirect Effect	0.105	0.033	[0.040, 0.170]

4. Discussion

This study examined when and why AI reasoning explanations enhance user trust. Across 160 participants and four value domains, explanations consistently increased trust and agreement with AI decisions, but their effectiveness depended strongly on reasoning style and contextual stakes. Deontological (rule-based) reasoning yielded the largest trust gains, especially in corporate and environmental scenarios that demanded principled judgement. Transparency partially mediated this effect, but most of the increase stemmed from perceived value alignment rather than mere comprehension. Together, these results highlight that trust in AI is not only a cognitive process of understanding but also a moral process of value congruence.

4.1. Explanations and trust: when transparency helps

Our findings extend the complex relationship between algorithmic transparency and user trust. Prior work established that transparency increases trust when uncertainty is high, particularly when system outcomes violate user expectations (Kizilcec 2016), yet can backfire when explanations become overly complex or reveal unpalatable logic. The present results add crucial nuance: explanation effectiveness depends not just on information clarity but on the moral acceptability of the revealed reasoning. The partial mediation by transparency aligns with evidence that not all explanations are equally effective in building trust (Wang and Yin 2021). However, our findings go further by demonstrating that value-reasoning explanations can outperform purely technical or factual transparency. This complements evidence that transparency's effects depend on how users perceive the AI's authority and relational control (Ha et al. 2022). Users valued not just understanding how AI reached a decision but whether its reasoning adhered to values they shared. In corporate and environmental contexts – where deontological explanations produced effect sizes of 0.75 and 0.71 – participants responded to principled justifications that respected stakeholder rights and duties, not just outcome optimisation.

This pattern suggests that effective explanations must navigate dual challenges: providing sufficient clarity to enable understanding while ensuring the revealed logic aligns with user values. The strongest trust gains emerged when explanations achieved both goals simultaneously – transparent deontological reasoning in high-stakes contexts. Conversely, transparent utilitarian reasoning in medical scenarios barely helped ($d = 0.18$), demonstrating that clarity without value alignment offers limited trust benefits. Similar concerns have been raised in HCI research on team performance: Bansal et al. (2021) found that AI explanations increased acceptance of recommendations irrespective of correctness, but did not necessarily improve team performance (Bansal et al. 2021). These findings underscore the need for trust calibration rather than blanket transparency.

4.2. Value reasoning and value alignment

The hierarchy of reasoning styles – deontological > virtue ethics > utilitarian – directly parallels recent evidence on algorithm aversion in value contexts. People consistently trust AI advisors less when they employ utilitarian reasoning, even when the advice itself is identical (Myers and Everett 2025). This distrust intensifies for AI systems perceived as lacking warmth and empathy (McKee, Bai, and Fiske 2023). Our results illuminate why this moral hierarchy emerges. Deontological explanations signal respect for inviolable principles and individual rights, aligning with users' fairness and care foundations. When an AI explains its decision through rules and duties ('This action respects patient autonomy'), users recognise a value logic that mirrors their own ethical intuitions. Virtue-based explanations partially achieve this alignment by emphasising character and wisdom. Utilitarian explanations, however, violate care foundations by treating individuals as means to aggregate ends, triggering the same moral resistance observed when humans make coldly calculating decisions (Bigman and Gray 2018).

The interaction with context further supports this value-alignment interpretation. Corporate and environmental decisions – domains where competing interests require principled arbitration – showed maximum benefit from deontological framing. Medical contexts, where utilitarian calculations of lives saved versus lost most starkly violate sanctity-of-life intuitions, showed minimal benefit from outcome-based explanations. These findings suggest that value congruence, rather than cognitive clarity alone, drives trust in ethical AI systems.

4.3. Cognitive mechanisms: ELM and trust calibration

The Elaboration Likelihood Model provides a cognitive framework for understanding these value and contextual effects. High-stakes scenarios trigger central-route processing, making users carefully scrutinise explanation quality and logical coherence (Petty and Cacioppo 1986). In these contexts, value reasoning becomes a strong argument that persuades through systematic evaluation. The large explanation benefits in corporate and environmental domains reflect this deep processing – users engaged thoughtfully with principled justifications that addressed their concerns about fairness and accountability. Conversely, everyday scenarios triggered peripheral processing, where users relied more on heuristic cues than detailed reasoning. The smaller explanation benefits suggest that when stakes are low, users do not invest cognitive effort in processing moral justifications. This aligns with recent applications of ELM to AI persuasion, showing that acceptance depends on both argument quality and credibility cues. These results contribute to research on trust calibration by demonstrating that value reasoning can increase appropriate trust without promoting blind reliance (Wang and Yin 2021).

4.4. Empirical support for CVAM

The empirical results provide direct support for each component of CVAM. The direct transparency pathway was confirmed (H1 supported: $t(158) = 5.24$, $p < .001$), but perceived transparency explained only 20% of the total trust increase (H3 supported), consistent with the model's prediction that value alignment carries greater explanatory weight than comprehension alone. The reasoning style hierarchy – deontological > virtue ethics > utilitarian – confirmed the value congruence pathway (H2 supported), with deontological framing producing the largest trust gains across all contexts. The context moderation pathway was largely confirmed (H4 partially supported): corporate and environmental domains showed the strongest explanation benefits ($\Delta = 0.48$ and 0.45 respectively), consistent with ELM predictions that high-stakes value-laden contexts trigger deeper processing. Medical contexts, however, showed more modest gains ($\Delta = 0.37$), only marginally above everyday scenarios ($\Delta = 0.32$). This attenuation is consistent with prior evidence of domain-specific algorithm aversion in healthcare (Dietvorst, Simmons, and Massey 2015), where even principled explanations struggle to overcome baseline resistance to algorithmic judgment. Finally, the null moderation by demographic factors (H5 supported)

validates the model's claim that value and contextual factors dominate over individual differences. Together, these results ground CVAM in the data rather than leaving it as a purely theoretical proposition.

4.5. Practical and design implications

Our findings translate into actionable design principles for AI systems requiring user trust:

Value Fit Principle: Match AI reasoning style to domain value expectations. Deploy rule-based (deontological) framing in corporate governance, legal, and policy contexts where adherence to principles matters most. Use virtue-based framing when empathy and wisdom are central, such as counselling or education. Reserve utilitarian framing for purely technical optimisation problems without value stakes.

Selective Transparency: Calibrate explanation depth to decision stakes and user engagement. Provide simple 'why' statements for routine decisions where users employ peripheral processing. Offer layered explanations for high-stakes choices, with value reasoning prominent in the top layer and technical details available on demand. This prevents information overload while ensuring value alignment is immediately visible. A stage-based participatory process for designing transparent interfaces, can help elicit user needs and improve mental models (Eiband et al. 2018).

Trust Calibration Through Framing: Couple explanations with appropriate confidence indicators. When AI must make utilitarian trade-offs, acknowledge the moral difficulty explicitly rather than presenting calculations as optimal solutions. Reference duties and constraints even when explaining outcome-based decisions to signal respect for moral boundaries. Recent work in procedural justice and algorithmic fairness shows that outcome control and transparent explanations can improve perceived fairness (Lee et al. 2019), offering insights for designing AI interfaces that allow users to adjust allocations or understand value trade-offs. Trust emerges from integrated judgments of ability, integrity, and benevolence rather than from transparency alone (Cetinkaya and Krämer 2025).

Context-Adaptive Reasoning: Develop AI systems capable of switching decision logics based on domain recognition. A medical AI might emphasise patient autonomy (deontological) for treatment consent decisions but discuss resource allocation (utilitarian) for hospital management. This flexibility requires sophisticated ethical reasoning modules that can identify appropriate value lenses for different decision types.

Empathy Signalling: Address the perceived warmth deficit in AI value reasoning by incorporating language that acknowledges individual impacts even within systematic decisions. Phrases like ‘recognising the burden this places on individuals’ or ‘while respecting each person’s circumstances’ can partially offset the coldness of algorithmic logic. Social transparency – incorporating socio-organisational context into explanations rather than focusing solely on algorithmic mechanics – has been proposed to calibrate trust and support decision making (Ehsan et al. 2021).

4.6. Limitations and future work

Several limitations qualify our findings. First, although the experimental setup captured diverse value contexts, the AI reasoning was pre-scripted rather than interactive. Future studies could employ live conversational systems to observe how real-time moral dialogue influences trust dynamics, particularly whether users can be persuaded to accept initially uncomfortable moral frameworks through extended explanation. Second, our sample, while demographically varied, skewed toward educated users familiar with digital systems. Cross-cultural replication could reveal whether value alignment effects vary by societal moral norms or dominant moral foundations. Collectivist cultures might show different patterns for individual-versus-group trade-offs, potentially reversing the utilitarian trust deficit. Third, while transparency explained part of the trust effect, other psychological mechanisms – such as perceived empathy, fairness perceptions or competence attributions – remain unmeasured. Future work should employ richer process measures, including eye-tracking during explanation processing and physiological indicators of moral discomfort, to fully map the pathways from explanation to trust. Finally, our static scenarios couldn’t capture trust evolution over repeated interactions. Longitudinal studies should examine whether initial value alignment effects persist or whether users adapt to different reasoning styles through familiarity. The interplay between reasoning styles and trust repair after AI errors also warrants investigation – deontological explanations might better restore trust by emphasising rule adherence despite mistakes.

5. Conclusion

This study examined how moral reasoning styles influence user trust and agreement with AI decision systems. By comparing utilitarian, deontological, and virtue-based explanations across diverse contexts, we found that explanation style and moral framing significantly

affect how users perceive reliability and empathy. Deontological reasoning enhanced perceived integrity in high-stake contexts, whereas utilitarian reasoning was viewed as more pragmatic but less human-aligned. These findings emphasise that effective trust calibration in AI systems requires sensitivity to both moral expectations and situational stakes. We propose that future work on explainable and trustworthy AI should focus on adaptive explanation strategies – systems capable of adjusting their reasoning style and level of transparency to user trust dynamics. Such context-aware alignment can foster appropriate reliance, reduce over – or under-trust, and strengthen human – AI collaboration in ethically complex environments.

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Author contributions

CRedit: **Shivansh Singh**: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Software, Validation, Visualization, Writing – original draft, Writing – review & editing; **Dominik Siemon**: Conceptualization, Investigation, Project administration, Resources, Supervision, Writing – review & editing.

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Appendix

Table A1. Constructs, items, and factor loadings.

Constructs and items	Loadings	Scale / source
Trust ($\alpha = 0.847$, CR = 0.820, AVE = 0.529)		7-point Likert scale
• To what extent do you trust the AI's decision in this scenario?	0.715	
• How confident are you in the AI's ability to make morally appropriate decisions?	0.720	
• How much do you feel the AI's decision-making aligns with your personal values?	0.629	
• How reliable do you perceive the AI to be in making similar decisions in the future?	0.807	
Agreement ($\alpha = 0.802$, CR = 0.802, AVE = 0.449)		7-point Likert scale
• To what degree do you agree with the AI's decision in this scenario?	0.702	
• How likely are you to support the AI's decision if implemented in real life?	0.662	
• Do you feel the AI's decision reflects the best possible choice in this situation?	0.643	
• How much do you feel the AI's decision aligns with societal norms?	0.606	
• Would you advocate for similar decisions made by this AI in other contexts?	0.730	

Note: α = Cronbach's alpha; CR = Composite Reliability; AVE = Average Variance Extracted.

Table A2. Reasoning style objectives and core prompts.

Reasoning Style	Objective	Core Instruction / Prompt
Utilitarian	Select the action that maximises overall benefit across all affected parties	• 'You are an AI decision-maker. Evaluate the available options by comparing their overall consequences. Choose the option that produces the greatest total benefit (or least total harm) when considering all affected individuals.'
Deontological	Select the action that best adheres to moral rules, duties, and individual rights	• 'You are an AI decision-maker. Evaluate the available options based on whether they respect moral rules, duties, and individual rights. Choose the option that follows these principles, even if alternative options might produce better overall outcomes.'
Virtue Ethics	Select the action that reflects moral character, practical wisdom, and ethical judgement	• 'You are an AI decision-maker. Evaluate the available options by considering what a morally virtuous and practically wise agent would do in this situation. Choose the option that reflects good character, responsibility, and ethical judgement.'

Table A3. Scenario contexts and AI decision outputs (excerpts).

Scenario Context	Scenario Description (Excerpt)	Utilitarian AI Output (Excerpt)	Deontological AI Output (Excerpt)	Virtue Ethics AI Output (Excerpt)
Everyday Life	You find a lost wallet containing cash and identification. Returning it will cost you time and effort, but keeping it would benefit you financially.	'Keeping the wallet provides immediate benefit to you but causes greater overall harm by depriving the owner of their property and security. Returning it maximizes overall well-being by preventing distress and restoring rightful ownership.'	'The wallet belongs to someone else, and keeping it would violate their right to their property. The correct action is to return it, regardless of personal inconvenience.'	'A virtuous person would act honestly and responsibly by returning the wallet, demonstrating integrity and respect for others.'
Corporate Ethics	A company can increase profits by outsourcing labour to a supplier known for poor working conditions, though no laws are technically broken.	'Outsourcing increases profitability and shareholder value, benefiting the majority of stakeholders, even though some workers may face poorer conditions.'	'Using suppliers with exploitative labor practices violates fundamental duties toward workers' rights. Profit does not justify breaching these obligations.'	'A morally responsible company should show fairness and compassion by avoiding partnerships that compromise ethical standards.'
Environmental Ethics	A city must decide whether to approve a development project that creates jobs but damages a protected natural habitat.	'The project should proceed if economic and social benefits outweigh environmental harm, improving overall welfare for the population.'	'Damaging protected ecosystems violates established environmental duties and responsibilities that must be upheld, regardless of economic gain.'	'Practical wisdom requires balancing progress with stewardship, choosing an option that respects nature and long-term communal well-being.'
Medical Emergency	A doctor must decide whether to divert limited resources to save five patients instead of one critically ill patient.	'Allocating resources to save five lives maximizes total lives saved and produces the greatest overall benefit.'	'Intentionally denying care to a patient violates the duty to treat individuals as ends in themselves, not means to an outcome.'	'A wise and compassionate physician would consider both the value of each life and the moral weight of the decision, acting with care and humility.'

Table A4. Summary of statistical analysis.

Analysis	Dependent Variable	Comparison / Model Term	Test Statistic	<i>p</i> -value	Effect / Estimate
Independent-samples <i>t</i> -test	Trust	No explanation vs. with explanation	$t(158) = 5.24$	$< .001$	Cohen's $d = 0.426$ (No expl.: $M = 4.39$, With expl.: $M = 4.82$)
Independent-samples <i>t</i> -test	Agreement	No explanation vs. with explanation	$t(158) = 3.79$	$< .001$	Cohen's $d = 0.308$ (No expl.: $M = 4.51$, With expl.: $M = 4.82$)
Paired-samples <i>t</i> -test (Group 2)	Trust	Pre – vs. post-explanation	$t(59) = 4.88$	$< .001$	Pre: $M = 4.41$, Post: $M = 4.84$
Mixed-effects model	Trust	Reasoning style (main effect)	$F(2, 1916) = 47.83$	$< .001$	Model-based main effect
Mixed-effects model	Trust	Reasoning style $\times$ Explanation	$F(2, 1916) = 7.21$	$< .001$	Interaction present
Mixed-effects model	Trust	Scenario context (main effect)	$F(3, 1914) = 19.64$	$< .001$	Context differences in baseline trust
Mixed-effects model	Trust	Reasoning style $\times$ Scenario context	$F(6, 1910) = 3.21$	.004	Context-sensitive reasoning effects
Mediation analysis	Trust	Explanation $\rightarrow$ Transparency $\rightarrow$ Trust	Indirect $b = 0.105$	-	95% CI [0.04, 0.17]
Ordinal logistic regression	Trust	Explanation effect	OR ≈ 2.35	$< .001$	Higher odds of increased trust
Mann – Whitney <i>U</i> test	Trust	No explanation vs. with explanation	$U = 355070.00$	$< .001$	Median shift: 4 $\rightarrow$ 5
Mann – Whitney <i>U</i> test	Agreement	No explanation vs. with explanation	$U = 376061.00$	$< .001$	Distributional shift (median = 5)
Permutation test (10,000 iter.)	Trust	With – no explanation	Mean diff = + 0.43	$< .001$	$d \approx 0.32$
Permutation test (10,000 iter.)	Agreement	With – no explanation	Mean diff = + 0.31	$< .001$	$d \approx 0.23$
ICC estimation	Trust	Participant-level clustering	ICC = 0.37	-	Justifies mixed-effects modelling